

MTN Proposal Mission Trajectory Data Package

1: Trajectory Information:

- Launch vehicle, launch date, C3, and departure launch azimuth
- Final orbit characteristics at Mars
- Binary spice SPK files or equivalent for beginning, middle, and end of launch window
- Solar conjunctions and eclipse durations
- Finite vs. Impulsive maneuver assumptions
- Aerobraking assumptions
- Delta-V budget

2: Spacecraft Data:

- Wet mass, propellant mass
- Chemical and/or electric engine types, number used, duty cycles, propellants, Isp, thrust and power characteristics (provide equations for thrust & power, Isp, mass flow rate)
- Thrust vectors for each maneuver
- Tank capacities and amount of propellant margin
- Power capability at 1 AU and amount of power margin
- Technology Readiness Levels and throughput capabilities of all engines, solar arrays, power processing electronics (PPU), etc.
- Momentum control method during quiescent cruise and during maneuvers